

Why the Moon Changes Shape: Modelling Lunar Phases

Javalab Phase of the Moon · Javalab

Year 7

~30 minutes

Science · Earth & Space Science

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Simulation: https://javalab.org/en/phase_of_moon_en/

Curriculum links: AC9S7U03

Learning intentions

- I can explain that the Moon's phase depends on its position relative to the Sun and Earth, not on Earth's shadow.
- I can predict the phase of the Moon at different points in its orbit.
- I can describe the pattern of lunar phases over one full cycle.

Getting started

Open the Javalab Phase of the Moon simulation. Before moving the Moon, look at the starting positions of the Sun, Earth and Moon, and note which phase is currently showing as seen from Earth.

Tasks

1. Move the Moon to the position directly between the Earth and the Sun. Predict the phase you'll see from Earth, then check the simulation.

2. Move the Moon to the opposite side of the Earth from the Sun. What phase is this, and what does it look like from Earth?

3. A classmate says the phases of the Moon are caused by Earth's shadow falling on it. Using what you observed, explain why this is incorrect.

4. Move the Moon to five different points around its orbit, predict the phase at each, then check against the simulation.

Moon's position in orbit	Predicted phase	Phase shown in simulation

5. Explain why an observer on Earth cannot see all of the Moon's phases on the same night.

6. Move the Moon slowly around one full orbit. How many distinct phases do you notice, and roughly how long does a full cycle take in real life?

7. Explain the relationship between the Moon's phase and how much of its sunlit side is facing Earth at that point in its orbit.

Extension (optional)

A lunar eclipse only happens during a full moon, but not every full moon produces one. Use what you know about the Moon's position and orbit to suggest why.